BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 092 · Compilers and Interpreters

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why is a translator needed for high-level source code?

Q2. Which type suits a count of pupils?
A. Real
B. Boolean
C. Character
D. Integer

Q3. “A character data type can hold an entire arbitrary-length sentence as one character.” Correct this. Explain why.

Q4. Start with unsigned 00000000. Set only the MSB to 1, then only the LSB to 1. Give both denary values. Why do they differ?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Translator — Program that converts source code into a form a computer can execute.

Compiler — Translator that processes the whole source program before execution and produces an executable.

Interpreter — Translator that executes a program as it translates it during running.

Source code — Instructions written by the programmer in a programming language.

Executable — A translated program that can be run by the target computer.

New learning

Why translation is needed: A processor runs machine instructions. A translator converts source code written in a programming language so the computer can execute it.

Compiler: A compiler translates the entire program before it is run and produces executable code. The executable can normally be run again without recompiling unchanged source.

Interpreter: An interpreter translates and executes the program during running, commonly one statement at a time. It usually has to be present each time the source is run.

Choose and compare: Compilation is useful when a finished program runs repeatedly or is distributed. Interpretation can be useful when developing and testing; error feedback can appear at the statement being executed.

Worked example

Explain TWO features of a compiler. [4 marks]

1. A programmer begins with source code: this is not normally directly executable by the CPU.

2. The compiler translates the whole source program before it runs [1], rather than executing each statement as it is processed [1].

3. The compiler produces executable machine code [1] that the CPU can run.

4. The executable can be run repeatedly without translating the unchanged source again [1].

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Q2. An interpreter translates and executes during running [1], allowing the learner to test changed code without a separate compilation step [1]. The interpreter is required at each run [1], which may make repeated execution slower than running a compiled executable [1].

Q3. The whole program is translated into executable code [1], so the distributed version can run on the target system [1]. The executable can run repeatedly [1], so recompilation of unchanged source is not needed each time [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A CPU runs machine instructions. High-level source code must be translated before or as it runs.

Q2. D — Integer

Q3. A character represents one character; a string represents a sequence of characters.

Q4. MSB only: 128. LSB only: 1. The difference is 127.

I Do / We Do pages · Guided practice

Q1. It translates and executes during running [1], allowing source code to be tested without producing a separate executable [1]. Errors can be reported at the statement reached [1], helping the programmer locate a fault [1].

Q2. A compiler translates the whole program before execution [1]; an interpreter translates and executes during running [1]. A compiled executable can normally run again without recompilation [1]; an interpreted program normally needs the interpreter each time [1].

Q3. The compiled executable can be distributed to run on the target platform [1], without each user needing the source-language interpreter [1]. The unchanged executable can be run many times [1], without repeating compilation on every launch [1].

You Do pages · Code First

```
print(ASC("B"))
```

Expected result: 66

You Do pages · Independent questions

Q1. The CPU needs executable machine instructions [1], so source code must be translated [1]. The compiler attempts to translate the whole source [1] and reports errors found during compilation so the programmer can correct them [1].

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Explain TWO reasons an interpreter might be useful while developing a program. [4 marks]

Worked method: It translates and executes during running [1], allowing source code to be tested without producing a separate executable [1]. Errors can be reported at the statement reached [1], helping the programmer locate a fault [1].

Guided 2

Compare when a compiler and an interpreter translate code, and what happens when the program runs again. [4 marks]

Worked method: A compiler translates the whole program before execution [1]; an interpreter translates and executes during running [1]. A compiled executable can normally run again without recompilation [1]; an interpreted program normally needs the interpreter each time [1].

Guided 3

An application is distributed to many school computers and run every day. Explain TWO benefits of compiling it before distribution. [4 marks]

Worked method: The compiled executable can be distributed to run on the target platform [1], without each user needing the source-language interpreter [1]. The unchanged executable can be run many times [1], without repeating compilation on every launch [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the missing closing bracket in this character-to-code conversion.

```
print(ASC("B")
```

iterate

Question 1

A compiler reports an error before the programmer can run a new program. Explain why translation is necessary and how the compiler helps find a problem. [4 marks] Source: Original OCR J277-style practice question.

Question 2

A learner changes the source code of a small program many times while testing. Explain ONE benefit and ONE drawback of interpreting the program. [4 marks] Source: Original OCR J277-style practice question.

Question 3

A game studio wants users to run a finished game repeatedly on their own computers. Explain TWO advantages of compiling the game. [4 marks] Source: Original OCR J277-style practice question.

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A compiler normally translates the whole source program before the executable runs.